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24k-0705 | BCS-4J
Data-Base System
Assignment # 01.

Question 1:

1. CyberNova Solution is facing several common problems associated with file based systems, these problems include data redundancy, data inconsistency, limited security, data isolation, poor data sharing leading to difficulty in retrieval of data.

2. By implementing a database management System, CyberNova solutions can store all data in a centralized and structured manner. A DBMS reduces data redundancy by maintaining a single integrated database, ensuring that updates are consistent within organization. It also improves data integrity by applying rules and validation, reducing errors. In addition, built-in security features such as user logins and access controls would protect sensitive customer and network information from unauthorized access.

3. A server based mobile application would allow employees in all branches to access the centralized database ~~anything~~ anytime and from anywhere. With real-time updates on customer details and service records, staff can handle issues more quickly and efficiently. Engineer and managers could use their mobile devices to stay connected, making their work more productive. Overall, the mobile app would improve communication, teamwork and daily operations of CyberNova Solutions.

~~Question 2~~

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Question 2:

1. Select * from clients where country = 'Pakistan';
2. Select service_name, Service_cost from Service where service_cost > 500;
3. Select * from service_records where status = 'completed';
4. Select client_name from clients where country = 'Pakistan' or country = 'United States'
5. Select * from service where service_cost between 300 and 1000;
6. Select DISTINCT location from offices;
7. Select * from service_records orders by service_date DESC;
8. Select UPPER (client_name) As client_name_uppercase from clients;
9. Select count(*) As total_service_records from Service_Records;
10. Select service_name, service_cost from services order by service_cost ASC;

Question 3:

1. Insert into Borrow_Records (Record_ID, Member_ID, Book_ID, Borrow_Date, Return_Date)
   Values (1004, 104, 5, '2026-02-05', NULL);
2. update Books set copies_Available = -1 where Book_ID = 2;
3. Delete from Members where Member_ID = 101;
4. Insert into Members (Member_ID, Member_Name, Email, Membership_Type)
   values (104, NULL, 'faiz@email.com', 'Premium');
5. Update Borrow_Records set Book_ID = 5 where Record_ID = 1002;
6. Insert into Books (Book_ID, Title, Author, Copies_Available)
   Values (4, 'Network Security', 'Ali Raza', 2);
7. Update Borrow_Records set Member_ID = 102 where Record_ID = 1003;
8. Delete from Books where Book_ID = 3;
9. Insert into Borrow_Records (Record_ID, Member_ID, Book_ID, Borrow_Date, Return_Date)
   values (1005, 102, 3, '2026-02-07', '2026-02-15')
10. Alter Table Books ADD constraint check_copies_available
    check (copies_Available >= 1);

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Question 4.

1. There are seven tables in the schema;
   - Book
   - Book_Authors
   - Publisher
   - Book_copies
   - Book_loans
   - library_Branch.
   - Borrower.

2. Book → BookID
   Book_Authors → BookID, AuthorName (composite key)
   Publisher → Name
   Book_copies → BookId, BranchId (composite key)
   Book_loans → BookId, BranchId, cardNo (composite key)
   Library_Branch → BranchId
   Borrower → Card No.

2. Book → PublisherName.
   - Tells which publisher published the book.

   Book_Authors → BookId.
   - Links authors to specific books,

   Book_copies → Book Id, BranchId
   - Shows how many copies of a book exist in a branch.

   Book_loans → BookId, BranchId, CardNo,
   - Tracks which borrower took which book from which branch.

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4.

Book:
- ↳ BookId (integer)
- ↳ Title (varchar)
- ↳ PublisherName (varchar).

Book_Authors:
- ↳ BookId (integer)
- ↳ AuthorName (varchar)

Publisher:
- ↳ Name (varchar)
- ↳ Address (varchar)
- ↳ Phone (varchar)

Book_loans:
- ↳ BookId (integer)
- ↳ BranchId (integer)
- ↳ CardNo (integer)
- ↳ DateOut (Date)
- ↳ DueDate (Date)

Book_Copies:
- ↳ BookId (integer)
- ↳ BranchId (integer).
- ↳ No_of_copies (integer)

Library_Branch:
- ↳ BranchId (integer)
- ↳ BranchName (varchar)
- ↳ Address (varchar).

Borrower:
- ↳ CardNo (integer)
- ↳ Name (varchar)
- ↳ Address (varchar)
- ↳ Phone (varchar).

5. This is a one-to-Many relationship between both. One book can have multiple authors. On the other hand each record in Book_Authors limited one author to one book. This is implemented using BookID as a foreign key in Book_Authors.

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Question 5:

1. Data independence is the ability of a database system to allow changes in the data structure without affecting application programs. It ensures that modifications at logical and physical level do not require changes in the programs that access the data. For example, if the storage method of a table changes from simple file structure to indexed storage, the application program can still use the same SQL queries without modification

2. A file based system can be better in situations where data is small. For example, a small shopkeeper maintaining daily sales in a text file as excel sheet does not need a full DBMS. A file system is cheaper, easier to manage and requires no special setup. In such cases, using a DBMS unnessary and expensive.

3. DELETE is used to remove specific rows from a table and can include WHERE to filter records. The table structure remains unchanged. TRUNCATE removes all rows from a table quickly but does not allow conditions. DROP completely removes the table from the database, including its structure and data.

4. Referential integrity ensures that relationships between tables remains ~~consti~~ consistent by enforcing valid foreign key references. It prevents the insertion of a foreign key value that do not exist in the referenced primary key table.

5. Yes, a foreign key can be NULL when the relationship is optional. This means the record does not have to reference another table. A NULL value does not violate referential integrity because it represents no relationship, not a valid one.

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